Bio-100 Assignment Forensic Biology

Name: _____

Student ID: _____

Case Title: Missing Kin- Clues Emerge in Aftermath of Attabad Disaster

In January 2010, a massive landslide hit the peaceful Hunza Valley in Northern Pakistan, reshaping the landscape and leading to the formation of Attabad Lake. The village of Attabad suffered significant damage, with 26 homes buried and 20 lives lost. Months later, as the waters of Attabad Lake receded, the remains of a submerged village were revealed. Two bodies surfaced among the recovered remains, raising the possibility that a boy from Karachi, missing since the landslide, might be among them. His parents in Karachi, who had lost contact with their son during the disaster, sought answers. The genetic material from the long-lost bodies became a crucial element in uncovering the mysteries hidden beneath the lake. The DNA results were sent to a forensic lab, where the grieving parents in Karachi also sent their blood samples.

The goal was clear: to analyse and compare the genetic signatures of the parents with those of the submerged victims in Attabad Lake. In pursuit of this objective, forensic scientists employed DNA fingerprinting analysis also known as DNA profiling. It is a sophisticated technique that identifies and analyses unique patterns within an individual's DNA. By examining specific regions of the genetic code, scientists can create a distinct genetic profile, similar to a fingerprint. This process helps establish genetic relationships and is crucial in forensic investigations by linking biological evidence to a specific individual. To initiate the process, scientists use restriction enzymes to precisely cut DNA segments. The resulting fragments are separated through gel electrophoresis, creating distinct banding patterns based on size and arrangement. These unique patterns, referred to as DNA profiles, serve as genetic signatures for each individual.

Task 1- DNA Fingerprinting

Case Data:

Using a fragment of DNA sequence (***150bp*** *long*) provided from one of the bodies, please answer the following questions.

5' **AAGGTATAGctcgagGGACACTGCAGGAAAAGCCACCTTGCGTTAGTTGTGAATTC**
ACAGAATTCGCTTGAGCCTCACTGCGAGTGGTTTTATTTGCTGCAGTTGCGAGAGTTTTGTC
GctcgagTTTCAAATAGGTGTATTTCTTCGAA **3'**

Below are the restriction enzyme recognition sites for ***XhoI***, ***PstI*** and ***EcoRI***. The place where the enzyme makes a cut has been marked with an arrow.

XhoI	PstI	EcoRI
▼	▼	▼
C TCGAG	C TGCAG	G AATTC

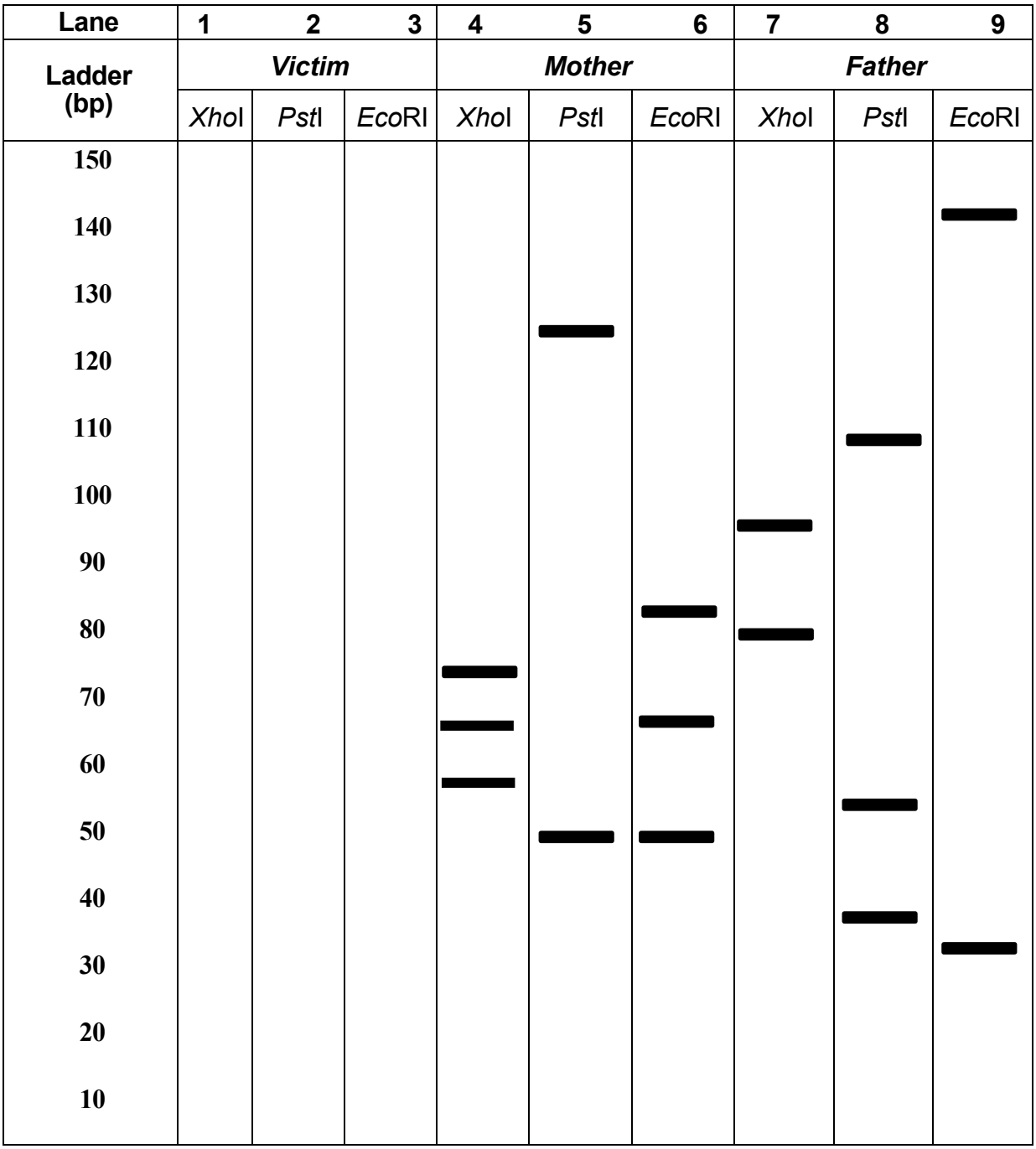
<u>Questions</u>

Q1. Identify and clearly label the **XhoI**, **PstI** and **EcoRI** restriction sites on the provided DNA sequence. (Use colour codes if needed!) **[6 marks]**

Q2. For each enzyme (XhoI, PstI and EcoRI), record the **number of cuts** induced in the DNA sequence, the **number of resulting fragments** and the **length of each fragment** in the table below. **[15 marks]**

Restriction Enzyme	Number of cuts	Number of fragments	Length of fragment (bp)
XhoI			
PstI			
EcoRI			

Q3. Use the data in the table above to draw banding patterns (on the provided gel image in **Lane 1, 2 and 3**) that would result from running the DNA fragments on an electrophoresis gel. **[9 marks]**



Q4. Compare the DNA fingerprint pattern belonging to the victim with that of the couple from Karachi- **Mother's** (**Lane 4,5 and 6**) and **Father's** (**Lane 7,8 and 9**) provided in the gel image above. Do you think the victim is their son? **[1 mark]**

Answer:

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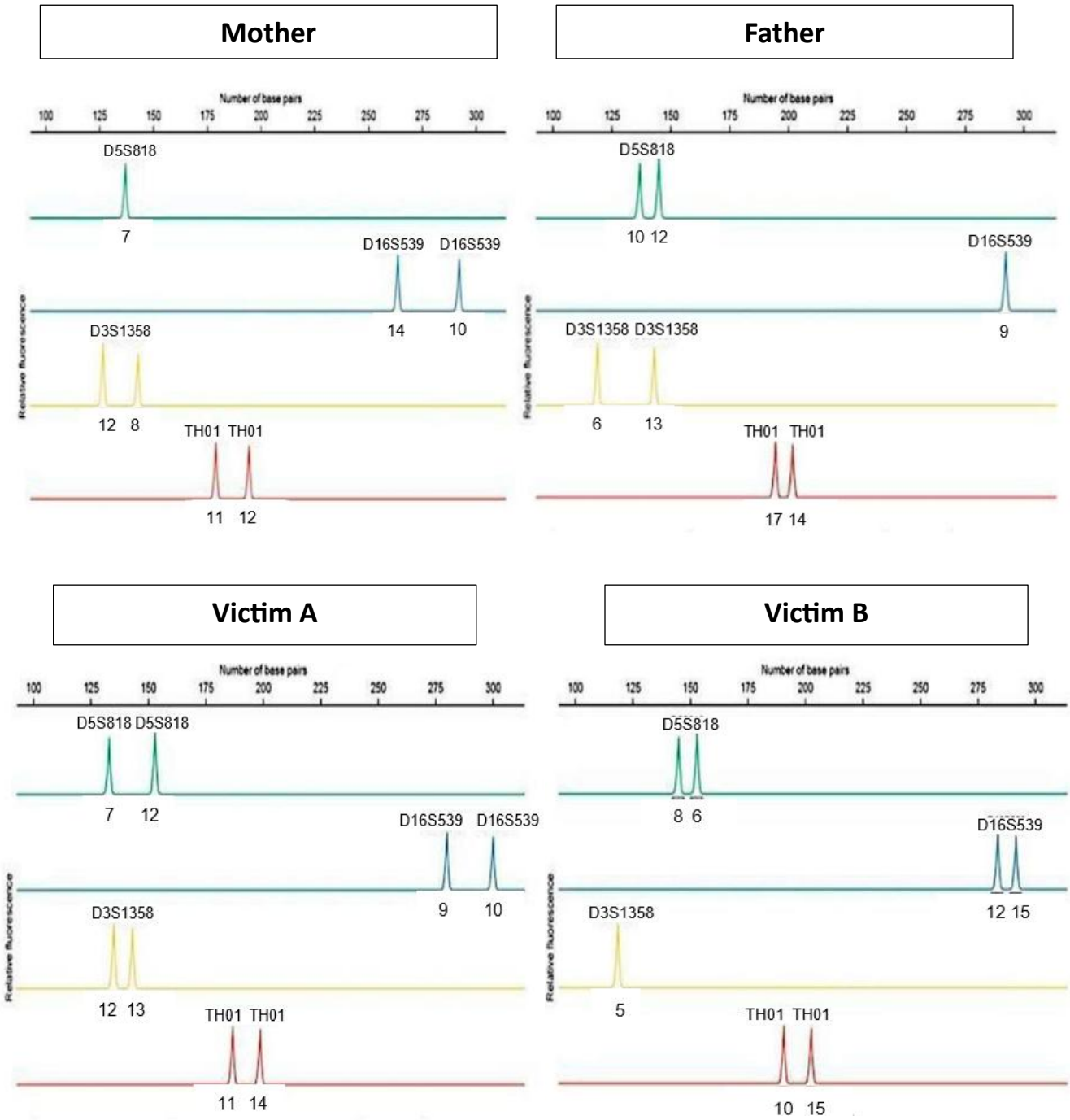
Task 2: STR Analysis

The provided DNA profiles depict the genetic information of the parents of the Karachi tourist and the two bodies recovered at the disaster site. Specifically, the information includes four *Short Tandem Repeats Loci* (STRs): **D5S818**, **D16S539**, **D3S1358**, and **TH01.** It's important to note that a comprehensive DNA profile would encompass additional STRs beyond the ones mentioned here.

Q1. Looking at the STR data below, which of the victims (**A** or **B**) seems to be the son of the Karachi parents? Use details from the figure below to explain your conclusion. *(Hint: Note that in offspring, one allele at each locus comes from the mother and one from the father.)* **[4 marks]**



Answer:

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